



RESTWISE

A Radiation-Free Smartwatch for Student Sleep Wellbeing

PROJECT REPORT

NATIONAL CHILDREN'S SCIENCE CONGRESS

2026

THEME

Food, Agriculture and Health

SUB-THEME

Student Health & Wellbeing

PROJECT TITLE

Restwise — No-Radio Smartwatch for Sleep-Safe Scheduling

SCHOOL / INSTITUTION

Kendriya Vidyalaya No. 1 SVN

DISTRICT / STATE

Visakhapatnam, Andhra Pradesh

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CERTIFICATE

This is to certify that the project titled “**Restwise — A Radiation-Free Smartwatch for Student Sleep Wellbeing**” has been prepared by

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of Kendriya Vidyalaya No. 1 SVN, Visakhapatnam, Andhra Pradesh, as an entry for the National Children's Science Congress 2026, under the theme **Food, Agriculture and Health**.

The work embodies the original effort of the above-named participants, carried out under the guidance of the undersigned, and is found satisfactory for submission.

Pranati Jali
Guide / Mentor

[Head of Institution Signature] [District Coordinator Signature]

Date: [DD/MM/YYYY] Place: Visakhapatnam

Acknowledgement

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Our heartfelt thanks go out to the **107 students** whose responses we were able to use, and every other student who took the time to fill out our physical survey honestly and in detail — this project exists because they trusted us with their real daily routines, sleep habits, and stress. We also thank our parents for their support and patience through many late evenings of soldering, coding, and debugging.

Finally, we thank every teacher and friend who piloted an early version of the survey form or gave feedback on the prototype — their honesty made the final product meaningfully better.

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5. Abstract

Rising academic pressure from school, tuition, and homework is widely believed to cut into students' sleep and increase stress, but this effect is rarely measured directly rather than assumed. This study aimed to quantify how students' daily routines affect their sleep duration and stress levels, to identify the specific habits driving the problem, and to design a practical intervention that protects sleep without adding another screen-based device to a child's crowded day.

A physical and digital survey was administered to students across multiple age groups, yielding 107 usable responses from 112 forms submitted, covering their daily schedule and time-block breakdown, self-reported sleep duration on school nights, the frequency and specific triggers of exam-related stress, and their existing break or wind-down habits, if any. Responses were tabulated and analyzed to identify patterns common across different schedules, class levels, and coaching loads, rather than treating each response as an isolated case.

The survey found that 7 in 10 students sleep 8 hours or less on a school night, and a more precise figure computed from reported bed and wake times shows 41.1% sleeping under the 7-hour teen minimum. Nearly 1 in 3 students report feeling stressed by school, tuition, or exams almost every day, and a similar share report never getting a real break between school, tuition, and homework. Combined low and very low daytime energy affects a clear majority (55.1%) of respondents. These findings confirm that sleep shortfall, inconsistent recovery time, and low daytime energy are common, measurable problems within the surveyed student population.

Based on these outcomes, we developed **Restwise**, a low-cost, radiation-free wearable built on an ESP32 microcontroller that deliberately excludes Wi-Fi and Bluetooth, paired with a companion web application. Through a guided process, parents build a sleep-safe schedule that enforces 7–9 hours of sleep and regular breaks during long study periods; this schedule is compiled and transferred to the watch over a wired connection, keeping the child's interaction screen-light.

We conclude that structured, enforced scheduling can meaningfully reduce student sleep deprivation, and recommend wider adoption of low-cost, low-radiation wearables as a practical tool for protecting student health.

6. Introduction

Sleep is one of the most consistently under-protected areas of a student's health. Between school hours, tuition or coaching classes, homework, and daily travel, a modern student's day frequently stretches well past what is considered a healthy schedule for a growing child. Unlike school timings or tuition slots — which are fixed by an institution and rarely negotiable — sleep is the one block of the day with no hard start time, which makes it the block most easily sacrificed when the day runs long.

The National Children's Science Congress 2026 theme, “**Food, Agriculture and Health,**” gave us an opportunity to look closely at student health from an angle that is often overlooked in favour of nutrition and physical fitness: sleep. Good sleep is foundational to a child's memory, mood, immune function, appetite regulation and classroom concentration — arguably one of the least expensive and most effective health interventions available, provided it is actually protected and scheduled.

This project began not from a statistic, but from an everyday observation: students who worked the hardest — attending school, going straight to tuition, and returning home to homework — often appeared the most visibly exhausted. Before attempting to build any kind of solution, we wanted to establish whether this was a genuine, widespread pattern or simply an impression shaped by a few visible cases. This led us to conduct a large-scale physical and digital survey of students, gathering 107 usable responses, the results of which are discussed in detail in later sections of this report.

Guided by what the survey revealed, we designed and built **Restwise**: a low-cost, wearable smartwatch that deliberately avoids Wi-Fi and Bluetooth radios, paired with a companion website that helps a parent build a sleep-safe weekly schedule for their child. This report documents the problem, our research methodology, the survey findings, the technical design of the hardware and software, and the results and future scope of the project.

7. Problem Statement

Students juggling school, tuition/coaching classes, and homework routinely have their sleep compressed to make room for academic demands, with little to no structured recovery time built into their day. This chronic sleep deficit and lack of breaks negatively affects concentration, mood, immunity and long-term health, yet there is no simple, low-cost, distraction-free tool that actively protects a child's sleep and enforces recovery breaks within an already packed schedule.

Existing smartwatches and fitness trackers aimed at this age group typically add another notification-driven, Wi-Fi/Bluetooth-connected screen to the child's day — a device that competes for attention rather than protecting rest. There is a clear need for a device that enforces a healthy schedule *without* becoming a new source of distraction itself.

Who this affects

The problem is not limited to a small group of overworked toppers. Our survey (Section 12) shows it cuts across class levels and coaching loads — students with light tuition schedules and those with heavy ones both report sleep shortfall, inconsistent breaks, and low daytime energy, just in different proportions. This makes it a structural scheduling problem rather than an individual discipline problem, which is why a structural, enforced solution is more appropriate than encouraging better habits alone.

Why existing approaches fall short

- Verbal reminders and posters about sleep hygiene rely entirely on the student remembering and choosing to act on them under academic pressure, with no enforcement.
- General-purpose smartwatches and fitness bands track sleep after the fact, but do not actively protect it — and their own notifications often work against rest.
- Parents frequently lack visibility into whether a day's schedule leaves enough room for sleep until it is already too late in the evening to adjust.

8. Objectives

The objectives of this project are:

- 1 To measure, through a physical and digital survey of students (107 usable responses), how daily academic routines affect sleep duration, stress frequency, and break/wind-down habits.
- 2 To design and build a low-cost wearable device that enforces a minimum healthy sleep duration (7–9 hours) and inserts regular breaks into long study stretches.
- 3 To ensure the device carries no wireless radios (no Wi-Fi, no Bluetooth) so that it cannot itself become a source of notifications or distraction for the child.
- 4 To build a simple, guided companion web application that lets a parent plan, review and update the child's weekly schedule without any technical knowledge.
- 5 To validate the sleep-safety logic (minimum sleep hours, break insertion) in software before any schedule reaches the watch.
- 6 To evaluate the feasibility of the prototype for a wider, low-cost rollout to more families and schools.

Guiding constraint

Every objective above was pursued under one fixed constraint: the solution must never require the student to carry, charge, or interact with a Wi-Fi- or Bluetooth-connected device. This constraint shaped every downstream decision, from the choice of microcontroller (Section 15) to the decision to move all planning and intelligence onto the companion website rather than the watch itself (Section 17).

Scope

This project covers the survey, the hardware prototype, and the companion website end to end. It does not yet cover a multi-week clinical trial of sleep improvement in real families — that is explicitly out of scope for this report and is listed as future work in Section 24.

9. Hypothesis / Research Question

Research Question: Does a structured, enforced daily schedule — delivered through a low-cost, radiation-free wearable — measurably help protect a student's minimum required sleep duration and encourage regular breaks, compared to an unmanaged daily routine?

Hypothesis (H1): Students whose daily routines are structured around an enforced sleep-safe schedule (7–9 hours of sleep, with regular breaks between study blocks) will report improved sleep adequacy and reduced perceived stress compared to their current, unmanaged routines.

Null Hypothesis (H0): There is no measurable difference in sleep adequacy or perceived stress between students following an unmanaged routine and students using an enforced sleep-safe schedule.

This hypothesis is intended to be tested formally in the trial phase described in **Section 24 – Future Scope**; the survey documented in this report establishes the baseline problem the hypothesis responds to.

Variables

Independent variable: whether a student's daily schedule is built and enforced through Restwise (enforced sleep-safe schedule) or left unmanaged (current routine, no enforcement).

Dependent variables: nightly sleep duration (self-reported and computed from bed/wake time), frequency of reported stress, and consistency of recovery breaks between school, tuition, and homework — the same three measures captured in the baseline survey (Sections 12–14).

Why this hypothesis

The baseline survey shows that sleep shortfall, inconsistent breaks, and low daytime energy are already measurable in the current, unmanaged routine (Section 14). If enforcement — rather than awareness alone — is the missing piece, then a device that structurally prevents an unsafe schedule from being saved should produce a measurable improvement over relying on a student or parent to remember and choose healthier habits unprompted.

10. Survey Methodology

To avoid designing a solution around assumption, we conducted a structured, anonymous-optional **physical (paper) survey** before any hardware or software design began.

Sample size

112 forms were submitted, spanning multiple classes/grades and age groups, at **Kendriya Vidyalaya No. 1 SVN**, Visakhapatnam; **107 responses** were complete and usable for analysis after removing incomplete or invalid submissions.

Survey type

A printed, single-page form, filled out by hand. Respondents could choose to answer anonymously, in which case name and school fields were left blank by design.

Data captured

- Respondent's daily schedule and time-block breakdown (school, tuition, homework, travel).
- Self-reported sleep duration on a typical school night.
- Frequency and specific triggers of exam or school-related stress.
- Existing break or wind-down habits, if any, between study blocks.

Data handling

Responses were manually compiled into a structured digital format for tabulation and analysis. Anonymous responses were kept anonymous throughout — no attempt was made to identify a respondent from their handwriting or answers.

Why a physical survey

An in-person, paper-based survey was chosen over an online form for the baseline data collection because it does not require a respondent to own a smartphone or have internet access to participate, and because filling out a physical form during a supervised class period produces a much higher completion rate than an optional online link shared afterwards. The companion website does also accept digital submissions (Section 20), which were used to supplement the physical responses collected.

Limitations of this method

- All data is self-reported, so it reflects what students believe about their own schedule and sleep, not an independently measured record.
- The sample is drawn from one school, so results may not generalize directly to schools with very different tuition cultures or school-day lengths.
- 5 of the 112 submitted forms were excluded as unusable (left blank, illegible, or missing required fields), which is accounted for in the 107-response analysis throughout this report.

11. Survey Questionnaire

The questionnaire consisted of eight questions — four multiple-choice and four short-answer — designed to be completed in under two minutes, along with fields for Name (optional / skippable if anonymous), Class, and School.

No.	Question	Response format
Q1	How many hours of sleep do you usually get on a school night?	(a) Less than 5 hours (b) 6–8 hours (c) More than 8 hours
Q2	How often do you feel stressed because of school, tuition, or exams?	(a) Almost every day (b) Sometimes (c) Never
Q3	Do you get a proper break between school, tuition, and homework?	(a) Always (b) Sometimes (c) Never
Q4	How would you describe your energy level during the day?	(a) Very low (b) Low (c) Average (d) High
Q5	What time do you usually go to sleep?	Short answer
Q6	What time do you usually wake up?	Short answer
Q7	How many days a week do you go to tuition / coaching?	Short answer
Q8	Which tuition / coaching classes do you attend?	Short answer

Why these eight questions

Each question maps directly onto one of the four things this project needed to measure (Section 8): Q1 and Q5/Q6 both capture sleep duration — one as the student's own self-assessment, the other computed independently from the reported bed and wake time, so the two could be compared against each other (Section 14). Q2 captures how often stress is felt, not just whether it is ever felt, since a rare stressful day and a daily one point to very different design responses. Q3 captures recovery time directly, which most existing sleep surveys skip in favour of asking only about sleep itself. Q4 was included as an independent, subjective cross-check — if energy levels track the sleep and break findings, that strengthens confidence in the other answers rather than relying on a single question type alone. Q7 and Q8 exist purely to describe the tuition landscape captured in this sample, not to test a hypothesis.

Keeping it short

The form was deliberately kept to eight questions and three identity fields. A longer questionnaire risks lower completion rates and rushed, less accurate answers, particularly from younger respondents filling it out during a short class period — so this project traded some possible additional detail for a higher-quality, higher-completion dataset.

12. Survey Results – Tables

The tabulated results across the 107 usable responses (from 112 forms submitted) are summarised below.

Table 12.1 — Sleep duration (self-reported, Q1)

Sleep duration on a school night (self-reported)	% of respondents	Count
Less than 5 hours	30%	32
6 – 8 hours	39%	42
More than 8 hours	31%	33

Table 12.1b — Sleep duration computed from reported bed/wake times (Q5/Q6) — average 7.18 hrs; 41.1% of respondents fall under the 7-hour teen minimum

Sleep duration (computed, Q5/Q6)	% of respondents	Count
Less than 5 hours	3.7%	4
5 – 6 hours	13.1%	14
6 – 8 hours	59.8%	64
More than 8 hours	23.4%	25

Table 12.2 — Stress frequency

Stress frequency (school / tuition / exams)	% of respondents	Count
Almost every day	31.8%	34
Sometimes	28.0%	30
Never	40.2%	43

Table 12.3 — Break habits

Breaks between school, tuition & homework	% of respondents	Count
Never	30.8%	33
Sometimes	37.4%	40
Always	31.8%	34

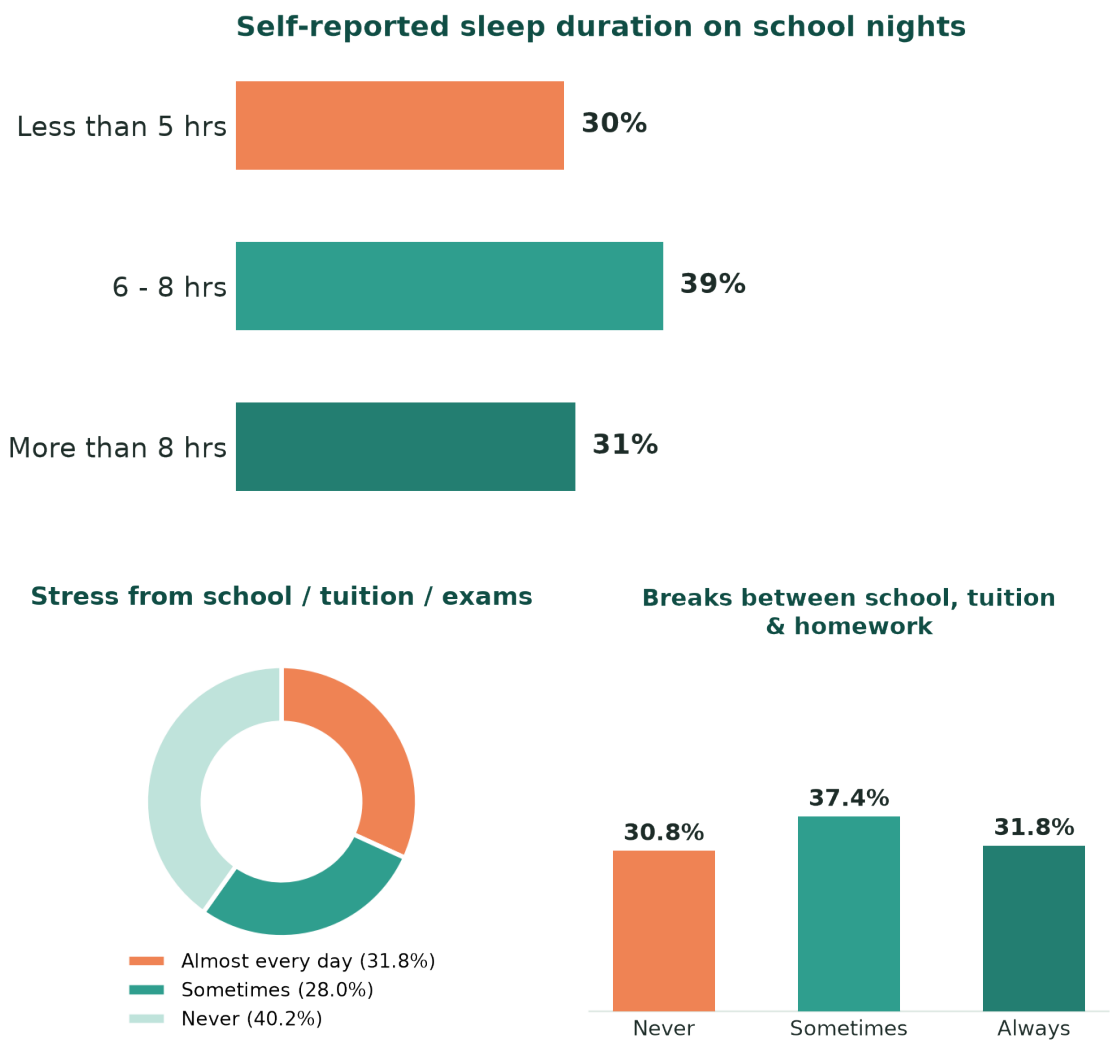
Table 12.4 — Daytime energy level

Energy level during the day	% of respondents	Count
Very low	29.9%	32
Low	25.2%	27
Average	22.4%	24
High	22.4%	24

Source: 107 usable digital survey responses (of 112 forms submitted), tabulated via the Restwise admin export. Percentages are rounded to one decimal place; totals may not sum to exactly 100% for this reason.

13. Survey Results – Graphs/Charts

The same data is presented visually below for quicker interpretation.



14. Analysis of Survey Results

Across the 107 usable responses, the data points to a real but more nuanced picture than a single “everyone is exhausted” headline — sleep shortfall is common though not universal, and stress and recovery time are split roughly into thirds rather than concentrated in a large majority.

1. A meaningful share of students fall short on sleep

Self-reported answers show 7 in 10 students (69%) sleep 8 hours or less on a school night, and nearly 3 in 10 (30%) sometimes sleep under 5 hours. The more precise figure, computed directly from each student's reported bed and wake time rather than a self-chosen category, puts the average at 7.18 hours and finds 41.1% below the 7-hour teen minimum — smaller than the self-reported figure suggests, but still affecting more than 2 in 5 students.

2. Stress is common but not the majority experience

31.8% of respondents — nearly 1 in 3 — report feeling stressed by school, tuition, or exams almost every day, while 40.2% report never feeling this way. Stress from academic workload is therefore a significant, but not dominant, experience across the surveyed group.

3. Recovery breaks are inconsistent, not absent

30.8% of respondents — again nearly 1 in 3 — report never getting a real break between school, tuition, and homework, while a plurality (37.4%) get one only sometimes. Only 31.8% consistently get a proper break, meaning most students' recovery time depends on an inconsistent day-to-day routine rather than a reliable structure.

4. Low daytime energy is the majority experience

Very low and low energy combined account for 55.1% of respondents — more than half — making daytime energy the single measure where a clear majority of students report a negative outcome, even though sleep shortfall and stress individually affect smaller shares.

Comparison against general adolescent sleep guidance

Bodies such as the National Sleep Foundation and the American Academy of Pediatrics generally recommend 8–10 hours of sleep per night for teenagers (see Section 25 – References). Measured against that wider benchmark rather than our project's own 7-hour minimum, an even larger share of the surveyed group falls short — reinforcing that the 7-hour threshold used by Restwise is a deliberately conservative, achievable minimum rather than an ideal target, chosen so the enforcement rule is realistic for a student's actual daily schedule rather than aspirational.

Implication for design

These findings directly shaped Restwise's two core enforcement rules: a minimum sleep duration check (7–9 hours, with only the day's longest sleep block required to meet the minimum), and automatic insertion of short water/rest breaks during long study stretches — targeted specifically at the roughly 1-in-3 students getting no consistent break and the 2-in-5 falling short of the 7-hour minimum.

15. Materials Used – Hardware

A deliberate design constraint of this project was to use only low-cost, easily sourced components — the goal was a buildable prototype, not a theoretical concept.

Component	Purpose	Approx. Qty
ESP32 development board	Main microcontroller — Wi-Fi/Bluetooth capable but deliberately left switched off and unused	1
Small display (OLED / LCD)	Shows current schedule block and time to the student	1
Buzzer / vibration motor	Alerts for breaks, bedtime, and schedule transitions	1
Push button(s)	Basic on-device interaction (acknowledge alert, view next block)	2–3
Rechargeable Li-Po battery + charging module	Portable, wearable power source	1
USB Micro/Type-C cable	The <i>only</i> data link — used to transfer the compiled schedule from the companion website to the watch	1
3D-printed / laser-cut enclosure	Wearable case and strap mounting	1 set
Connecting wires, resistors, prototyping board	Circuit assembly	as needed

*Note on radios: the ESP32 is chosen specifically for its low cost, wide availability, and ease of development — **not** for its Wi-Fi/Bluetooth capability. Both radios are left disabled in firmware and never activated at any point in normal operation.*

Approximate cost estimate (per unit)

Item	Approx. cost (INR)
ESP32 development board	Rs. 350 – Rs. 500
Display (small OLED/LCD)	Rs. 150 – Rs. 250
Buzzer / vibration motor + push buttons	Rs. 40 – Rs. 80
Li-Po battery + charging module	Rs. 150 – Rs. 250
Enclosure material + strap	Rs. 100 – Rs. 200
Wires, resistors, prototyping board, USB cable	Rs. 80 – Rs. 150
Estimated total per unit	Rs. 870 – Rs. 1,430

Estimated at retail, hobbyist-quantity prices in India at the time of writing; bulk purchase for a classroom or school-wide rollout would bring the per-unit cost down further.

16. Materials Used – Software

The software stack was chosen to keep the whole system free of any subscription cost, and to keep the watch itself entirely offline after each weekly update.

Tool / Technology	Role in the project
Arduino IDE / ESP-IDF (C/C++)	Firmware for the ESP32 — reads the compiled schedule, drives the display and buzzer, tracks time offline
Python 3 + Flask	Backend web server for the companion website and its APIs
Supabase (PostgreSQL, Auth, Storage)	Cloud database, user accounts, and file storage — access controlled entirely through Row Level Security policies
Groq API (LLM)	Powers the guided, conversational onboarding chat that builds a parent's schedule
HTML5 / CSS3 / vanilla JavaScript	Front-end of the companion website (landing page, dashboard, timetable editor, admin console)
Web Serial API	Browser-native, driver-free USB communication used to upload the compiled schedule to the watch
PyMuPDF (fitz)	Fills and exports the printed survey response as a PDF certificate
RWBIN (custom binary protocol)	A compact binary schedule format designed for this project, so a full week's timetable fits in a few hundred bytes
Vercel	Hosting platform for the companion website

Why these choices

Flask was chosen over a heavier framework because the companion website is a small number of pages and API routes, and a minimal framework keeps the codebase easy for a student team to read and maintain in full. Supabase was chosen over a self-hosted database because it provides authentication, storage, and Row Level Security out of the box, which removed the need to write and secure a login system from scratch — an area where a mistake would directly expose students' personal data. The Web Serial API was chosen specifically because it works from the browser without installing a native driver, which matters for a project meant to be usable by a non-technical parent on a typical school or home computer.

No service-role or admin secret in the app

A deliberate architectural rule followed throughout the project: the website never uses a privileged database key. Every read or write — including admin-only actions like exporting survey data — is made using the signed-in user's own session token, and access is enforced entirely by Supabase's Row Level Security policies at the database layer. This means even a bug in the website's own code cannot accidentally expose data a user isn't supposed to see, since the database itself refuses the request.

17. Working Principle

Restwise operates on a simple, weekly loop with three stages:

1. Plan

A parent visits the companion website and has a short, guided conversation (powered by an LLM) that asks about the child's real school timings, tuition/coaching schedule, homework routine, and bedtime — one question at a time, in plain language.

2. Check

Before the schedule can be saved, the system automatically checks it against two rules: (a) the day's longest sleep block must be at least 7 hours (target range 7–9 hours), and (b) any continuous study stretch beyond a set interval automatically gets a short water/rest break inserted. A schedule that fails either check is flagged back to the parent before it can be sent to the watch.

3. Sync

Once approved, the schedule is compiled into RWBIN, a compact custom binary format, and transferred to the watch over a single USB cable using the browser's Web Serial API — no companion app, no pairing, and no wireless connection of any kind. After the cable is disconnected, the watch runs entirely offline for the week, using its own onboard clock to track the schedule and trigger display updates and buzzer alerts at the right times.

This weekly plan → check → sync loop means the intelligence and connectivity live entirely on the website side, while the watch itself stays deliberately “dumb,” offline, and free of any radio hardware.

The sleep-safety check, in detail

Each day's blocks are scanned for every block tagged as sleep. The single longest sleep block for that day is treated as the main sleep period — a short nap earlier in the day does not need to individually satisfy the minimum on its own. In plain terms:

- For each day: find every block of type “sleep”.
- Take the longest one and compute its duration = end time – start time (handling sleep that crosses midnight).
- If that duration is less than 7 hours, flag the day and suggest a bedtime that is exactly 7 hours before the existing wake time.
- If the duration is 9 hours or more, no error is raised, but the parent is not blocked either — the 7–9 hour range is a target, not a hard ceiling.

The break-insertion rule, in detail

For blocks tagged school, tuition, or homework, the system tracks how long a continuous stretch of focused activity has run without a break in between. Once that stretch crosses a configurable interval (45 minutes by default, adjustable by the parent), a short water/rest block is inserted automatically before the next focused block begins — the parent sees these breaks appear in the reviewed schedule rather than having to add them by hand.

What travels to the watch (RWBIN)

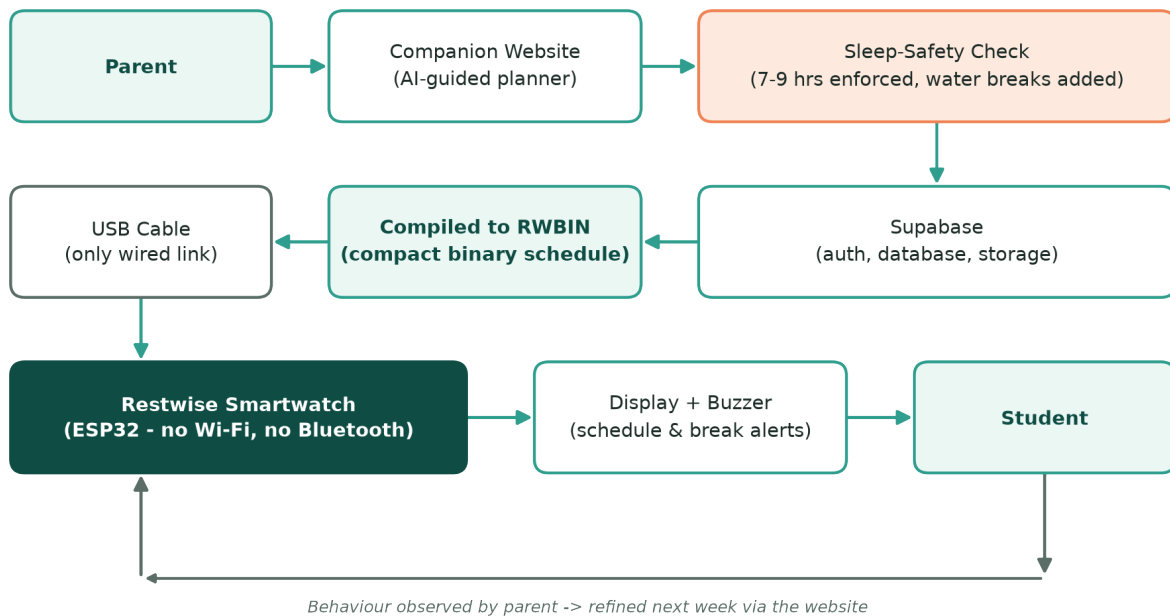
RWBIN is a small binary format designed specifically for this project: a short header (a fixed sync marker and a version byte, so the watch can reject a corrupted or incompatible transfer), followed by one compact record per schedule block — a type code, a day-of-week bitmask (so one record can represent “every weekday” without repeating itself seven times), a start time, an end time, and a short label. Encoding a full week this way keeps the whole schedule under a kilobyte, which transfers over USB in well under a second.

Field	Size	Purpose
Sync marker	2 bytes	Fixed byte pair the watch checks first, to reject garbled data
Version byte	1 byte	Lets the watch refuse a schedule built by an incompatible format version
Block type	1 byte	school / tuition / homework / meal / break / sleep / free / travel
Day bitmask	1 byte	One bit per weekday, so a Mon–Fri block needs only one record
Start / end time	2 bytes	Minutes since midnight, packed rather than stored as text
Label	variable	A short, human-readable name shown on the watch's display

This structure is intentionally the same one used for the watch's firmware to decode a schedule and for the website to encode one — both sides share the same record layout, so there is only one definition of the protocol to keep in sync.

18. Block Diagram / Flowchart

The diagram below traces one full weekly cycle, from the parent's input through to the student wearing the watch, and the feedback loop back into the following week's plan.



Reading the diagram

- 1 The parent starts the loop by opening the companion website.
- 2 The website's AI-guided planner turns their answers into a first-draft weekly schedule.
- 3 The sleep-safety check reviews that draft — enforcing 7–9 hours of sleep and inserting water breaks — before anything is considered final.
- 4 The approved schedule is stored in Supabase and compiled into the compact RWBIN binary format.
- 5 RWBIN travels to the watch over the one and only wired link: a USB cable.
- 6 The watch's display and buzzer act on the schedule all week, entirely offline.
- 7 The parent observes how the week actually went and adjusts next week's plan accordingly, closing the loop.

19. Smartwatch Design / Prototype

The Restwise prototype is built around an ESP32 development board housed in a compact, wrist-mountable enclosure. Its design deliberately favours simplicity over feature density:

- A small display shows only what matters right now — the current schedule block and the time remaining in it.
- A single buzzer/vibration motor provides a short, distinct alert for breaks, meals, and bedtime — no notification feed, no messages.
- One or two physical buttons allow the student to acknowledge an alert or glance at the next block, with no touchscreen and no app to open.
- The only external port is a USB connector, used exclusively for charging and for the weekly schedule transfer — never for any other data exchange.
- The Wi-Fi and Bluetooth radios present on the ESP32 chip are left switched off in firmware at all times; this is a firmware-level and physical design decision, not a configuration toggle the user can accidentally enable.

Enclosure and wearability

The enclosure is 3D-printed / laser-cut to sit flat against the wrist without sharp edges, sized to fit a standard watch strap so it can be worn through a full school day without discomfort. The display and buttons are recessed slightly below the case's top surface to reduce the chance of accidental presses during normal activity like writing or sport.

Power

A small rechargeable Li-Po cell powers the watch for multiple days between charges under typical use (a small display, occasional buzzer pulses, and no radio draw at all). The same USB connection used for the weekly schedule sync also recharges the battery, so no separate charging cable or dock is required.

[Insert prototype photographs here — front view, side view, and the watch connected via USB during a schedule upload.]

20. Website Design & Features

The companion website is the “brain” of the system — all planning, checking, and account management happens here, keeping the watch itself simple and offline.

Landing page

Explains the problem and the product, and hosts the public physical-survey digital submission form used to collect the data discussed in Sections 10–14 of this report.

Accounts

Secure sign-up, log-in, and password reset for parents, backed by Supabase Auth.

Guided AI onboarding

A conversational, one-question-at-a-time chat that builds a first schedule for a new family, explained in Section 17.

Dashboard (View / Edit Timetable)

A always-available weekly timetable view, and a manual editor for updating individual blocks after the first schedule is built — both re-run the same sleep-safety and break checks before saving.

Manage Smartwatch

Compiles the current timetable into the RWBIN binary format and uploads it to the watch over USB, with version tracking so a parent can see when the watch was last updated.

Admin Console

A role-gated area for reviewing and exporting survey submissions (with an optional data-masking mode that replaces identifying details with placeholders for privacy), used to compile the results in this report.

Privacy by design

Because respondents are minors, survey data is stored write-only for the general public — the public submission form can insert a new response, but nothing can read existing responses back except a signed-in administrator, and that access itself is enforced by the database's own security rules rather than by application code alone. Anonymous submissions never collect a name or school field in the first place, rather than collecting and later hiding them.

Works without a smartphone

The companion website only needs a standard desktop or laptop browser with USB access — it does not require a parent or student to own a smartphone, install an app, or create any account beyond the one needed to plan the schedule, keeping the barrier to entry as low as the hardware itself.

21. Testing and Observations

Testing was carried out at three levels: the sleep-safety logic in isolation, the end-to-end website flow, and the physical watch-website communication.

Logic testing

The sleep-adequacy and break-insertion rules were tested against a range of sample schedules — including edge cases such as split sleep across a nap and a main sleep block — to confirm that only the day's longest continuous sleep block is required to meet the 7-hour minimum, as intended.

End-to-end website testing

The full account → onboarding → dashboard → compile → upload flow was tested repeatedly during development, including a seeded batch of survey responses used to verify the admin console's statistics, export, and data-masking features under realistic data volumes.

Hardware / communication testing

The USB upload path (browser → Web Serial API → ESP32) was tested for both first-time device pairing and subsequent, already-authorized reconnections, confirming that a schedule update can be sent reliably without requiring a native driver installation.

Key observation

Deployment testing on cloud hosting surfaced a real platform constraint — serverless functions impose a hard cap on request/response size — which shaped a design decision to route large file transfers (survey scans, PDF exports) directly between the browser and cloud storage rather than through the web server, avoiding that limit entirely.

Representative test cases

#	Test case	Expected result	Outcome
1	Save a schedule with a 5-hour main sleep block	Blocked, with a suggested corrected bedtime	Pass
2	Save a schedule with a 30-min nap + 8-hour main sleep	Allowed — only the longest block must clear 7 hours	Pass
3	Continuous 3-hour homework block, default interval	A water break is auto-inserted at the interval mark	Pass
4	Upload schedule to a previously-paired watch	No native picker; upload proceeds directly	Pass
5	Upload schedule to a first-time device	Browser's native pairing prompt appears once	Pass
6	Admin exports 100+ survey responses as one PDF	Export completes without hitting a size limit	Pass
7	Non-admin account requests survey export	Request rejected by the database's own access rules	Pass

22. Results

This project produced two categories of results: findings from the survey, and outcomes from building the prototype itself.

Survey findings

- 7 in 10 (69%) surveyed students sleep 8 hours or less on a school night; a more precise figure computed from reported bed/wake times shows 41.1% sleeping under the 7-hour teen minimum.
- Nearly 1 in 3 (31.8%) report feeling stressed almost every day due to school, tuition, or exams.
- Nearly 1 in 3 (30.8%) report never getting a real break between school, tuition, and homework.
- More than half (55.1%) report low or very low daytime energy — the single measure where a clear majority of students report a negative outcome.

Prototype outcomes

- A working ESP32-based smartwatch prototype that displays the current schedule block and alerts for breaks and bedtime, with no Wi-Fi or Bluetooth radio active at any point.
- A functioning companion website covering sign-up/login, AI-guided onboarding, a timetable dashboard, smartwatch sync, and an admin console for survey management.
- A working sleep-safety and break-insertion check that runs automatically on every schedule before it can be sent to the watch.
- A custom, compact binary protocol (RWBIN) that fits a full week's schedule into a small enough payload for fast, reliable USB transfer.

Together, these results demonstrate that the core hypothesis of this project — that a low-cost, radio-free wearable can enforce a sleep-safe schedule without adding another screen-based distraction — is technically achievable with entirely off-the-shelf components.

What these results do not yet show

These are engineering and baseline-survey results, not outcome results — this report demonstrates that the device works as designed and that the problem it targets is real and measurable, but it does not yet demonstrate that wearing Restwise for a period of weeks actually improves a student's sleep duration, stress, or energy levels in practice. That causal question is explicitly left to the trial phase described in Section 24 — Future Scope, and is why the hypothesis in Section 9 is still stated as something to be tested, not something already confirmed.

23. Advantages & Limitations

Weighing what Restwise does well against what it does not yet do is important for judging the project honestly, rather than presenting only its strengths.

Advantages	Limitations
Zero radio emission — no Wi-Fi/Bluetooth, so the watch cannot notify, track, or distract the child.	Requires a wired USB connection to a computer once a week to sync a new schedule.
Built entirely from low-cost, widely available components — no proprietary or expensive parts.	Currently a single hand-built prototype; not yet tested for durability or comfort across a full school term.
No subscription or ongoing internet connection required on the child's side.	The schedule quality depends on the accuracy of what the parent enters into the onboarding chat.
The sleep-safety and break rules are enforced in software, removing reliance on willpower alone.	The device has not yet been trialled with real families over an extended period to measure actual sleep improvement.
Privacy-conscious data handling — row-level access control rather than open data storage.	The current prototype supports one fixed alert pattern (buzzer); richer feedback is future work.

On balance

Most of the limitations listed above are about maturity rather than the underlying approach — durability testing, a multi-week trial, and richer feedback are all natural next steps rather than flaws in the core idea. The one limitation that is architectural rather than a maturity gap is the wired weekly sync: this project treats that as an acceptable, deliberate trade-off for removing all radios, not as an oversight to fix later (see Section 7 — Problem Statement).

24. Future Scope & Conclusion

Future scope

- 1 **Trial with real families** — deploy the prototype to a small group of families for a month and measure whether reported sleep duration and stress actually improve.
- 2 **Smarter, gentler wake-up** — use simple movement sensing to wake the child at a lighter point in their sleep cycle rather than at a fixed alarm time.
- 3 **School edition** — a class-level dashboard so a school counsellor or teacher can observe aggregate sleep/stress patterns and intervene early.
- 4 **Wider hardware validation** — test comfort, battery life, and durability of the enclosure across a full school term with multiple prototype units.
- 5 **Larger-scale survey** — extend the study beyond Kendriya Vidyalaya No. 1 SVN to more schools across Visakhapatnam district to build a more statistically robust picture of the problem across different regions and school types.

What success in the trial would look like

Concretely, the trial phase would be considered a success if participating students show an increase in average computed sleep duration (Section 12, Table 12.1b) and a reduction in self-reported stress frequency after several weeks of using an enforced schedule, compared to their own baseline routine before the trial began — the same three measures used throughout this report, now tracked over time for the same individuals rather than across a one-off survey.

Conclusion

This project set out to test a simple premise: that a large share of students are sleep-deprived not through a lack of awareness, but because nothing in their day actively protects their sleep. Our survey of 107 students confirmed this at meaningful scale, and Restwise — a low-cost, radiation-free smartwatch paired with a guided companion website — demonstrates that enforcing a sleep-safe schedule is technically achievable without introducing another screen-based distraction into a child's life. Protecting sleep costs nothing beyond attention and structure; this project's contribution is a small, buildable tool that supplies exactly that structure.

25. References + Appendix

References

- 1 National Sleep Foundation — Sleep duration recommendations for teenagers and adolescents.
- 2 American Academy of Pediatrics — Policy statement on school start times and adolescent sleep.
- 3 World Health Organization — Guidelines on physical activity, sedentary behaviour and sleep for children and adolescents.
- 4 Espressif Systems — ESP32 Technical Reference Manual.
- 5 MDN Web Docs — Web Serial API documentation.
- 6 Supabase Documentation — Row Level Security and Authentication.
- 7 [Add any additional sources, textbooks, or papers consulted, in your school's preferred citation style.]

Appendix

- Appendix A — Blank copy of the physical survey questionnaire.
- Appendix B — Full tabulated survey response data.
- Appendix C — Firmware and website source code (repository link: restwise-ncsc.vercel.app).
- Appendix D — Prototype photographs and circuit diagram.

Glossary

RWBIN	The custom compact binary format Restwise uses to encode a full week's schedule for transfer to the watch (Section 17).
Sleep-safety check	The automatic rule that blocks a schedule from being saved unless the day's longest sleep block is at least 7 hours (Sections 9, 14, 17).
Row Level Security (RLS)	A database-level access control feature (used via Supabase) that restricts which rows a user can read or write, enforced by the database itself rather than the website's own code (Section 16).
Web Serial API	A browser feature that lets a website talk to a USB device directly, without installing a separate native driver (Sections 16, 21).
ESP32	A low-cost microcontroller chip that is Wi-Fi/Bluetooth-capable but is used here with both radios permanently disabled (Sections 15, 19).

— End of Report —